「Commutator

「Def. Let G be a group. Given $x, y \in G$, define

$$[x, y] \overset{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the Commutator of x and y. And, given two subset A and B,

$$[A, B] \overset{\text{def}}{=} \langle [a, b] \mid a \in A,\ b \in B \rangle$$

Particularly, $G' \overset{\text{def}}{=} [G, G]$ is called the derived set.

「Prop. G' char G, (So $G' \trianglelefteq G$).

Proof. Given $\sigma \in Aut(G)$ and $[x, y]$ for arbitrary $x, y \in G$,

$$\sigma([x,y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

So, $\sigma[[G, G]] = [G, G]$.

「Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H, G] \leq H$

Proof. $H \trianglelefteq G \iff$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$[H, G] \leq H \iff$ for all $g \in G$ and $h \in H$, $[h, g] = h^{-1}g^{-1}hg \in H$.

So proved.

「Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x, y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$.

Note: G' is the smallest normal subgroup of G s.t. the factor is abelian.

Note: If G is abelian, then $G' = 1$.

Note: $[x, y] \cdot [y, z] = x^{-1}y^{-1}xy\,y^{-1}z^{-1}yz = x^{-1}y^{-1}xz^{-1}yz$

$$= \left(\begin{array}{l} [x, y]^{-1} = y^{-1}x^{-1}yx = [y, x] \\ [x, y^{-1}] = x^{-1}yxy^{-1} \\ [x^{-1}, y] = xy^{-1}x^{-1}y. \\ [x, y][y, x] = x^{-1}y^{-1}xy\,y^{-1}x^{-1}yx = id. \end{array} \right.$$

「Commutator of Typical Subgroups.

「D_{2n}, the dihedral of order $2n$.

ΓS_n, the symmetric group on n letters

$[S_3, S_3] = \langle [a,b] \mid a, b \in S_3 \rangle$.

$[(1\,2),(1\,3)] = (1\,2)(1\,3)(1\,2)(1\,3) = (1\,3\,2)(1\,3\,2) = (1\,2\,3)$

$[(1\,2),(2\,3)] = (1\,2)(2\,3)(1\,2)(2\,3) = (2\,3\,1)(2\,3\,1) = (1\,3\,2)$

$[(1\,3),(2\,3)] = (1\,3)(2\,3)(1\,3)(2\,3) = (3\,2\,1)(3\,2\,1) = (1\,2\,3)$

$[(1\,2),(1\,2\,3)] = (1\,2)(1\,2\,3)^{-1}(1\,2)(1\,2\,3) = (1\,2)(3\,2\,1)(1\,2)(1\,2\,3) = (1\,3\,2)$

$[(1\,3),(1\,2\,3)] = (1\,3)(1\,2\,3)^{-1}(1\,3)(1\,2\,3) = (1\,3)(3\,2\,1)(1\,3)(1\,2\,3) = (1\,3\,2)$

$[(2\,3),(1\,2\,3)] = (2\,3)(1\,2\,3)^{-1}(2\,3)(1\,2\,3) = (2\,3)(3\,2\,1)(2\,3)(1\,2\,3) = (1\,3\,2)$.

$[(1\,2\,3),(1\,3\,2)] = id$.

Therefore, $[S_3, S_3] = \langle (1\,2\,3), (1\,3\,2) \rangle = \{id, (1\,2\,3), (1\,3\,2)\} = A_3$.

Alternatively, Since A_3 is normal subgroup of S_3 and $S_3/A_4 \cong C_2$, so $[S_3, S_3] \subseteq A_3$. Conversely, since $(1\,3\,2) \in A_3$ and $(1\,3\,2) = [(1\,2),(2\,3)] \in [S_3, S_3]$, Therefore $[S_3, S_3] \supseteq A_3 = \langle (1\,3\,2) \rangle$.

A_n, the alternating group.

ΓQ_8. The Quaternion.